THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET A

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of radius 7 cm and angle 60° is: 1

(a) $\frac{154}{3}$ cm² **(b)** 77 cm² **(c)** 22 cm² **(d)** $\frac{77}{3}$ cm²

2. The length of an arc of radius 21 cm subtending 120° is: 1

(a) 44 cm **(b)** 132 cm **(c)** 66 cm **(d)** 22 cm

3. The area swept by a 7 cm minute hand in 15 minutes is: 1

(a) 38.5 cm² **(b)** 11 cm² **(c)** 154 cm² **(d)** 77 cm²

4. A student states: "The length of a semicircular arc is πr^2." The correct response is: 1

(a) Correct **(b)** It is $2\pi r$

(c) Incorrect — it is πr **(d)** Cannot be determined

5. A sector has area one-fourth of the circle. Student A says its angle is 90°; Student B says 45°. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. The volume of a hemisphere of radius 3 cm is: 1

(a) 9π cm³ **(b)** 36π cm³ **(c)** 18π cm³ **(d)** 27π cm³

7. The curved surface area of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 1540 cm² **(b)** 440 cm² **(c)** 308 cm² **(d)** 220 cm²

8. Two solid hemispheres of radius r are joined base to base. The surface area of the solid is: 1

(a) $4\pi r^2$ **(b)** $2\pi r^2$ **(c)** $3\pi r^2$ **(d)** $6\pi r^2$

9. A student states: "The surface area of a toy made of a cone on a hemisphere is $\pi rl + 3\pi r^2$." The correct response is: 1

(a) It is $\pi rl + \pi r^2$ **(b)** Incorrect — it is $\pi rl + 2\pi r^2$

(c) Correct **(d)** Cannot be determined

10. For the volume of a cone of radius 7 cm and height 12 cm, Student A says 616 cm³; Student B says 1848 cm³. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of radius 12 cm with angle 30°. (Use $\pi = 3.14$) 2

12. The circumference of a circle is 88 cm. Find the length of an arc subtending 90° at the centre. 2

13. A cylinder of radius 3.5 cm and height 12 cm is surmounted by a cone of the same radius and height 12 cm. Find the volume of the solid. 2

14. Two cubes of edge 10 cm are joined end to end. Find the surface area of the cuboid formed. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the minor segment of a circle of radius 20 cm if the angle of the sector is 60°. (Use $\pi = 3.14$, $\sqrt{3} = 1.73$) 3

16. A solid is a cone of height 8 cm mounted on a cylinder of height 10 cm, both of radius 6 cm. Find its total surface area. 3

17. A circular park of radius 21 m has a sector-shaped pond of angle 60°. Find the area of the pond and the remaining area of the park. 3

OR

In a circle of radius 6 cm, a sector has area 18.84 cm². Find its angle and arc length. (Use $\pi = 3.14$) 3

18. A vessel is a hollow hemisphere of radius 3.5 cm surmounted by a hollow cylinder of the same radius and height 8 cm. Find its capacity. 3

OR

A toy is a cone of radius 4 cm and height 3 cm mounted on a hemisphere of radius 4 cm. Find its surface area. (Use $\pi = 3.14$) 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A pizza of radius 14 cm is cut into 8 equal slices. 5

(i) Find the area of each slice.

(ii) Find the length of the crust (arc) of each slice.

(iii) If 3 slices are eaten, find the area of pizza left.

OR

A horse is tied to a corner of a square field of side 40 m by a 21 m rope. 5

(i) Find the area it can graze.

(ii) Find the area it cannot graze.

(iii) Find the increase in grazing area if the rope is 28 m.

20. **Case study:** A water bottle is a cylinder of radius 4 cm and height 15 cm with a hemispherical top. (Use $\pi = 3.14$) 5

(i) Find the volume of the cylindrical part.

(ii) Find the volume of the top.

(iii) Find the capacity in litres.

(iv) Find the curved area of the cylinder covered by a label.

OR

A rocket model is a cylinder of radius 3 cm and height 12 cm with a cone of radius 3 cm and height 4 cm on top. Find 5

(i) its volume

(ii) its total surface area.

END OF QUESTION PAPER — **Total: 40 Marks**